



Dr. Hidetaka Kuniyoshi (he/him)

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Current Position

1. Japan Society for the Promotion of Science (JSPS) Overseas Research Fellow (Apr. 2025-)
Host institution: Solar and Space Physics Group, Northumbria University, UK

Education

1. **PhD (awarded 24th March 2025)**,
Dpt. of Earth and Planetary Science, The University of Tokyo (Apr. 2022–Mar. 2025)
Thesis: [The role of swirls in solar coronal heating](#)
2. **MSc (awarded 24th March 2022)**
Dpt. of Earth and Planetary Science, The University of Tokyo (Apr. 2020–Mar. 2022)
3. **BSc (awarded 24th March 2020)**
Dpt. of Earth and Planetary Physics, The University of Tokyo (Apr. 2016–Mar. 2020)

Fellowships, Awards & Grants

1. International Astronomical Union (IAU) PhD Prize 2025
2. Japan Society for the Promotion of Science (JSPS) Overseas Postdoctoral Research Fellowships, April 2025–March 2027 (£76,000)
3. The University of Tokyo School of Science Research Award (Doctor, ranked 1st in Dpt. Of Earth and Planetary Science), March 2025
4. The University of Tokyo School of Science Research Award (Master), March 2025
5. International Travel Support for Students, Institute for Space-Earth Environmental Research (ISEE), Nagoya University, June 2023 (£980)
6. Bronze Medal, Japan Physics Olympiad, August 2014

Publications

First-author Papers

1. **Kuniyoshi, H.**, & Morton, R. J., *A self-consistent solar coronal heating model by Alfvénic waves*, submitted to ApJ.
2. **Kuniyoshi, H.**, Imada, S., Yokoyama, T., 2025, *A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation*, ApJL, 990, L71.
5 times cited ([NASA/ADS](#)): Top 10% most cited paper in Space Science (Web of Science).
3. **Kuniyoshi, H.**, Bose, S., Yokoyama, T., 2024, *Comprehensive Synthesis of Magnetic Tornado: Cospacial Incidence of Chromospheric Swirls and Extreme-ultraviolet Brightening*, ApJL, 969, L34.
15 times cited ([NASA/ADS](#)): Top 10% most cited paper in Space Science (Web of Science).

4. **Kuniyoshi, H.**, Morton, R. J., Shoda, M., Yokoyama, T., 2024, *Can the Solar p-modes Contribute to the High-frequency Transverse Oscillations of Spicules?*, *ApJ*, 960, 118. 8 times cited ([NASA/ADS](#)).
5. **Kuniyoshi, H.**, Shoda, M., Iijima, H., Yokoyama, T., 2023, *Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer*, *ApJ*, 949, 8. 28 times cited ([NASA/ADS](#)): Top 10% most cited paper in Space Science (Web of Science).
6. **Kuniyoshi, H.**, Hesse, M., Norgren, C., Tenfjord, P., Kwagala, N. K., 2021, *Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing*, *JGR: Space Physics*, 126, e29236. 1 time cited ([NASA/ADS](#))

Co-author Paper

7. Silva, S. S. A., et al. (incl. **Kuniyoshi, H.**), *Observing solar vortices with existing and future instrumentation. Solar Physics International Network for Swirls (SPINS) white paper (Helio)*, White paper submitted to UK Space Frontiers 2035 on 28 ([NASA/ADS](#))
8. Baker, D., et al. (incl. **Kuniyoshi, H.**), 2024, *Searching for Evidence of Subchromospheric Magnetic Reconnection on the Sun*, *ApJ*, 970, 39. 6 times cited ([NASA/ADS](#)).

Oral Presentations at Conferences and Workshops (6 invited talks)

2026

1. Coronal Loop Workshop XII, “A self-consistent solar coronal heating model by Alfvenic waves”, Newcastle, UK, 10 Jun. 2026
2. Kyoto University A-KDK Supercomputer Symposium, “A self-consistent solar coronal heating model by Alfvenic waves”, Online, 05 Mar. 2026
3. International Conference on “Exploring the Sun at High Resolution: Present Perspectives & Future Horizons”, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Udaipur, India, 10-13 Feb. 2026 (**invited**)
4. Center for Computational Astrophysics (CfCA) Users Meeting, “A self-consistent solar coronal heating model by Alfvenic waves”, Tokyo, Japan, 23 Jan. 2026

2025

5. Rironkon Symposium, “A self-consistent solar coronal heating model by Alfvenic waves”, Ibaraki, Japan, 17–19 Dec. 2025
6. Waves and Instability in the Solar Atmosphere (WISA), “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Rome, Italy, 23-26 Sep. 2025
7. Hinode18–IRIS16 Meeting, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, London, UK, 23–27 Jun. 2025 (**invited**)
8. Solar Physics International Network for Swirls (SPINS) Meeting, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Sheffield, UK, 2–4 Jun. 2025 (**invited**)
9. Astronomical Society of Japan (ASJ) Spring Meeting, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Ibaraki, Japan, 17–20 Mar. 2025
10. Japan Solar Physics Community (JSPC) Symposium, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Kanagawa, Japan, 17–19 Feb. 2025 (**invited**)
11. Cool Stars Workshop in Japan, “A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.”, Tokyo, Japan, 12–14 Feb. 2025

2024

12. Rironkon Symposium, "A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.", Tokyo, Japan, 24–26 Dec. 2024
13. ASJ Fall Meeting, "Comprehensive Synthesis of Magnetic Tornado: Cospacial Incidence of Chromospheric Swirls and Extreme-ultraviolet Brightening", Hyogo, Japan, 11–13 Sep. 2024
14. Hinode-17 IRIS-15 SPHERE-3 Meeting, "Comprehensive Synthesis of Magnetic Tornado: Cospacial Incidence of Chromospheric Swirls and Extreme-ultraviolet Brightening", Montana, USA, 23–27 Jul. 2024
15. Research Institute for Mathematical Sciences (RIMS) Symposium "Non-equilibrium Turbulence", "Comprehensive Synthesis of Magnetic Tornado: Cospacial Incidence of Chromospheric Swirls and Extreme-ultraviolet Brightening", Kyoto, Japan, 22–24 Jul. 2024 (**invited**)
16. ASJ Spring Meeting, "Quiet Sun small-scale brightening initiated by a magnetic tornado", Tokyo, Japan, 11–15 Mar. 2024
17. Solar-C SWG Meeting, "Quiet Sun small-scale brightening initiated by a magnetic tornado", Nagoya, Japan, 5–6 Mar. 2024
18. JSPC Symposium, "Quiet Sun small-scale brightening initiated by a magnetic tornado", Tokyo, Japan, 20–22 Feb. 2024
19. Chromospheric Magnetic Field Workshop, "Quiet Sun small-scale brightening initiated by a magnetic tornado", Tokyo, Japan, 14–16 Feb. 2024 (**invited**)
20. CfCA Users Meeting, "Quiet Sun small-scale brightening initiated by a magnetic tornado", Tokyo, Japan, 29–30 Jan. 2024

2023

21. Hinode-16 / IRIS-13 Meeting, "Can the Solar p-modes Contribute to the High-frequency Transverse Oscillations of Spicules?", Niigata, Japan, Sep. 2023
22. ASJ Fall Meeting, Nagoya, "Can the Solar p-modes Contribute to the High-frequency Transverse Oscillations of Spicules?", Japan, Sep. 2023
23. WISA 2023, "Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer", Newcastle, UK, Jun. 2023
24. ASJ Spring Meeting, "Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer", Tokyo, Japan, Mar. 2023
25. JSPC Symposium, "Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer", Nagoya, Japan, Feb. 2023

2022

26. ASJ Fall Meeting, "MBP merger, an alternative solar/stellar coronal heating mechanism" Niigata, Japan, Sep. 2022
27. Japan Society of Fluid Mechanics (JSFM) Annual Meeting, "MBP merger, an alternative solar/stellar coronal heating mechanism", Kyoto, Japan, Sep. 2022
28. SOLAR MHD Conference, "MBP merger, an alternative solar/stellar coronal heating mechanism", Eastbourne, UK, Aug. 2022
29. Japan Geophysical Union (JpGU) Meeting, "MBP merger, an alternative solar/stellar coronal heating mechanism", Chiba, Japan (hybrid), May 2022
30. JSPC Symposium, "MBP merger, an alternative solar/stellar coronal heating mechanism", Online, Feb. 2022
31. American Geophysical Union (AGU) Fall Meeting, "MBP merger, an alternative solar/stellar coronal heating mechanism", New Orleans, USA (hybrid), Dec. 2021
32. JpGU Meeting, "Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing", Online, May 2021

33. ASJ Spring Meeting, "Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing", Online, Mar. 2021

Poster presentations at Conferences and Workshops

2026

1. Cool Stars 23, Ariake, "A self-consistent solar coronal heating model by Alfvénic waves", Japan, 15-16 Jun. 2026

2025

2. International Astronomical Union (IAU) Symposium 400, "A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.", Medellín, Colombia, 21–25 Jul. 2025
3. National Astronomical Meeting (NAM), "A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.", Durham, UK, 7–11 Jul. 2025

2024

4. CfCA Users Meeting, "A unified picture of swirl-driven coronal heating: magnetic energy supply and dissipation.", Tokyo, Japan, 25–26 Nov. 2024

2023

5. Rironkon Symposium, "Quiet Sun small-scale brightening initiated by a magnetic tornado", Aomori, Japan, Dec. 2023
6. AGU Fall Meeting, "Magnetic Tornado Properties: A Substantial Contribution to the Solar Coronal Heating via Efficient Energy Transfer", San Francisco, USA, Dec. 2023
7. JpGU Meeting, "Can the Solar p-modes Contribute to the High-frequency Transverse Oscillations of Spicules?", Chiba, Japan, May 2023

2022

8. IAU General Assembly, "MBP merger, an alternative solar/stellar coronal heating mechanism", Busan, ROK (hybrid), Aug. 2022

2020

9. AGU Fall Meeting, "Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing", Online, Dec. 2020
10. JSPC Symposium, "Magnetic Reconnection in a Sheared Magnetic Flux Tube: Slippage Versus Tearing", Online, Dec. 2020

Invited Seminar Talks

1. Plasma Dynamics Group (PDG) seminar, Sheffield University, UK, 09 Oct. 2025
2. Solar Orbiter Atmospheric Heating WG Seminar (online), 5 May 2025
3. Yamada Lab Seminar, Waseda U, Japan, 9 May 2025
4. Seminar, Nagoya U (invited), Aichi, Japan, Jul. 2024
5. Seminar, Tsukuba Space Forum, U. Tsukuba (invited), Ibaraki, Japan, Apr. 2024
6. Seminar, Dept. of Earth Science & Astronomy, U. Tokyo (invited), Tokyo, Japan, Oct. 2023
7. Seminar, Mullard Space Science Laboratory, UCL (invited, online), Aug. 2023
8. Seminar, Plasma Dynamics Group, U. Sheffield (invited, online), May 2023

High Performance Computer (HPC) and Observational experience as PI

HPC

1. **ATERUI III**, *12 million CPU hours*, Centre for Computational Astrophysics (CfCA) in National Astronomical Observatory of Japan (NAOJ), Japan, 2026
2. **A-KDK**, *1.7 million CPU hours*, Research Institute for Sustainable Humanosphere (RISH) in Kyoto Univ., Japan, 2026
3. **DiRAC**, *2.8 million CPU hours*, Science and Technology Facilities Council (STFC), UK, 2026
4. **ATERUI III**, *12 million CPU hours*, CfCA in NAOJ, Japan, 2025
5. **A-KDK**, *1.7 million CPU hours*, RISH in Kyoto Univ., Japan, 2025
6. **ATERUI II**, *12 million CPU hours*, CfCA in NAOJ, Japan, 2024
7. **A-KDK**, *1.7 million CPU hours*, RISH in Kyoto Univ., Japan, 2024

Observation

8. **Daniel K. Inouye Solar Telescope (DKIST)**, *3.2 hours*, DKIST Cycle 4, 2026

Conference organizing experiences

1. **Session Chair**, “Linking Plasma Dynamics in the Lower Solar Atmosphere to Magnetic Activity”, Asia Oceania Geoscience Society, Aug. 2026 (accepted),
2. **LOC**, Coronal Loop Workshop XII, Northumbria University, Jun. 2026